A Joint Siamese Attention-Aware Network for Vehicle Object Tracking in Satellite Videos

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***Abstract*— Remote sensing object tracking (RSOT) is a novel and challenging problem due to the negative effects of weak features and background noise. In this article, from the perspective of attention-focus deep learning, we propose a joint Siamese attention-aware network (JSANet) for efficient remote sensing tracking which contains both the self-attention and cross-attention modules. First, the self-attention modules we propose emphasize the interdependent channel-wise coefficient via channel attention and conduct corresponding space transformation of spatial domain information with spatial attention. Second, the cross-attention is designed to aggregate rich contextual interdependencies between the Siamese branches via channel attention and excavate association produces reliable correspondence with spatial attention. In addition, a composite feature combine strategy is designed to fuse multiple attention features. Experimental results on the Jilin-1 satellite video datasets demonstrate that the proposed JSANet achieves state-of-the-art performance in terms of precision and success rate, demonstrating the effectiveness of the proposed methods.**

***Index Terms*— Attention mechanism, satellite videos, Siamese tracker, vehicle object tracking.**

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I. Introduction

REMOTE sensing imagery, compared with natural imagery, has the characteristics of broader coverage area and higher maneuverability, which endow it with incomparable advantages in the fields of disaster assessment [1], urban monitoring [2], and resource exploration [3]. Especially in recent years, with further technology development of orbital static and attitude transformation, several remote sensing video satellites which are capable of monitoring upon the specific area and providing tight temporal association images stream continuously have been successfully launched, such as SkySat [4], TUBSAT [5], and Jinlin-1 [6], which not only realized the leapfrog development from image to video but also give rise to a new research area: remote sensing object tracking (RSOT) [7]–[10].

With the increasing popularity of research, more and more remote sensing tracking algorithms have been proposed. It is worth noting that these methods are often improved from the natural tracking field. Therefore, how to construct a paradigm realizing efficient transformation between nature and remote sensing methods has become a hot field issue. The core point of the transfer paradigm is "What are the core points of RSOT"?

Take the Jilin-1 video frame shown in Fig. 1 as an example. Remote sensing data inevitably have the disadvantage of ultralow resolution, negative impacts of illumination, atmospheric particles, and clouds during space imagery further reduce the imaging quality. Therefore, the RSOT methods need to focus more on how to solve the feature scarcity [11]–[13] and clutter disturbance [14], [15] problems. In addition to overcoming the influence of fuzzy noise, and defining the boundary between the target and the background, more consideration should also be given to designing effective correlation strategies, discovering the interrelationships between the template and the search feature [8], [16], [17].

Attention mechanism [18]–[20] is a data-processing strategy inspired by the physiological mechanism of the human brain. Lock areas require critical focus and invest more visual resources to get more detailed information meanwhile suppress other useless information. The cross-attention [21] is capable of aggregating rich contextual interdependencies between the template and the search image, enabling the search branch to generate a more discriminative representation, helping identify

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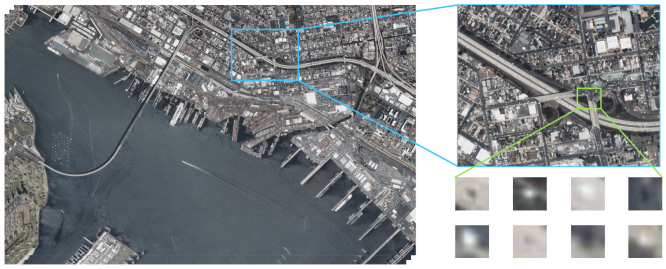


Fig. 1. Example Jilin-1 video frame. The blue-boxed region in the left side image is enlarged, and then the vehicles in the green-boxed region are further enlarged. The pixel numbers of these targets are generally less than 80, which are typically weak targets. 1) Contain less appearance, sometimes like a mosaic. 2) Easy to be mixed with other objects or distractions. 3) Blur with background, difficult to delimit boundary line.

the target more accurately. Meanwhile, target representation can be more meaningful when the contextual information of the search image is encoded. It is well worth trying to consider the advantages of an attention mechanism well-suited to the RSOT problem.

At present, some scholars have proposed some attention trackers [21], [22]. After collection and analysis, the existing methods still have the following shortcomings.

1) Poorly self-attention design. The structural design of channel and spatial modules has strong convergence, ignoring the difference between spatial and channel attention.
2) Lack of cross-spatial mechanism. This is mostly because the sizes of attention maps in spatial dimensions are inconsistent and lack an implementation basis for interaction.

In this work, we propose a joint Siamese attention-aware network (JSANet), apart from verifying the performance of Siamese attention trackers meanwhile ameliorating insufficient existing methods. In view of the above two problems, the main contributions of this work can be summarized as follows.

1) Propose a new Siamese attention tracker, through joint optimization of multiple attention modules, implementing information screening and focusing on key areas, improving the robustness against weak features and background noise.
2) Design a self-attention that contains both channel and spatial modules, focus more on the channels with higher scores, conduct corresponding space transformation of spatial domain information, and achieve precise targeting of core areas.
3) Propose cross-attention, aggregate meaningful contextual interdependencies between the target template and the search image via cross-channel attention (CCA), and effectively excavate the correlation between objects with temporal association.

The rest of this article is organized as follows: Section II introduces the preliminaries of related trackers and attention models. Section III elaborates the proposed method in detail. Section IV first introduces the characteristics of data used and the experimental details, and then shows the experimental results and the corresponding analysis on the five datasets. Finally, Section V draws the conclusion of this article.

II. Related Work

In this section, we will introduce the contents of two aspects. First, the existing object tracking methods in the remote sensing field. Second, the attention mechanism as the core of this article is described in detail.

A. Object Tracking in Remote Sensing

Compared with the relatively clear developing venation in natural object tracking, scholars in the remote sensing field are still trying to solve the problem with diversified directions. For instance, naive Bayes [23], background subtraction [24], target modeling [25], optical flow [26], and so on.

In addition to the above methods, at present, there is a recessive tendency in the latest development of RSOT. That is, adding targeted strategies based on a classical correlation filter algorithm to get a more suitable remote sensing tracking model. For instance, velocity correlation filter (VCF) [27] uses both velocity feature and inertia mechanism proposed with kernelized correlation filter (KCF) [56]. Deep hyperspectral KCF (DeepHKCF) [28] applies popular concepts from conventional object trackers and combines a self-designed ROI mapping strategy. Likelihood of features tracking with kernelized correlation (KC-LoFT) [29] adds a decision module using peak-to-sidelobe ratio into the KCF module to prevent error accumulation from blending nontarget regions to the target template during an update. Correlation filter with motion estimation (CFME) [9] combines the Kalman filter and motion trajectory averaging to mitigate the boundary effects, solving the problem of tracking failure when a moving object is partially or completely occluded.

The reason why the correlation filter algorithms represented by KCF are favored by many researchers is mainly because of their low computational complexity, high running speed, and fixed feature extraction mode without repeated iterative learning.

B. Self-Attention Mechanism

Deeply influenced by the idea of duality [30], the self-attention models [21], [31], [32] can be further subdivided into self-channel and self-spatial submodules to be calculated. The channel attention emphasizes the channel-wise relevance coefficient between channels and key information, guiding the network to focus more on the channels with higher scores. Spatial attention conducts corresponding space transformation of spatial domain information and further identifies core areas that deserve more attention which contributes to extracting the key information more efficiently. After the summary and analysis of the models that use this pattern, their structural design of them usually has strong convergence. In [21], after introducing the structure of the self-spatial, self-channel is simply expressed as "We can compute channel self-attention in a similar manner." In addition to structural convergence, what rather important is ideological convergence: self-channel deeply influenced by SENet and self-spatial by convolutional block attention module (CBAM) [33]. Although these follow-on models all have innovative amelioration, however, the root problems of [20], [33] have not been well-solved: the

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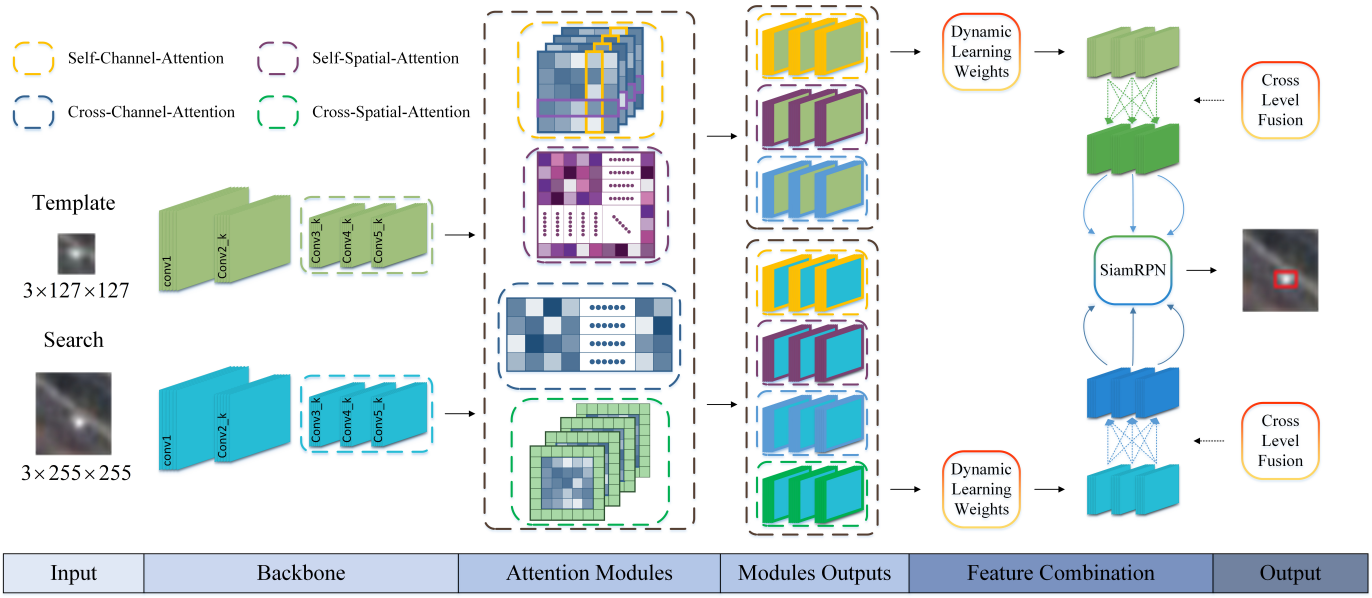


Fig. 2. Overview of the proposed tracker JSANet. It consists of four types of attention modules, a feature combination module, and Siamese region proposal networks (SiamRPNs). The features of the last three stages are extracted and then modulated by attention and feature combination modules. It generates two-stream attentional features which are fed into SiamRPN blocks to predict a single tracking region. As shown in the upper left corner, we use different colors' dotted boxes to indicate the identity of each attention module. The same colors are used in the "Modules Outputs" part to indicate the relationships between output features and attention modules.

dimension reduction will destroy the direct correspondence between the channel and its weight.

III. Joint Siamese Attention-Aware Network

In this section, the modules in JSANet are explained and analyzed in detail. As illustrated in Fig. 2, it consists of two main components: attention modules and a feature combination module. In the following subsections, we will first revisit the Siamese-based trackers in Section III-A to provide a clear description of the proposed method, and then the design ideas and detailed description of self-attention, cross-attention, and feature combination modules are successively introduced in Sections III-B–III-D.

A. Siamese-Based Trackers

Bertinetto *et al.* [34] first introduced Siamese architecture into the tracking domain, namely, SiamFC, which takes an image pair as input, comprising an exemplar image z and a candidate search image x. Both the inputs are processed by a convolutional network φ with parameters θ. The similarity function uses cross correlation as formula as

$$f_\theta(z,x) = \varphi_\theta(z) \star \varphi_\theta(x) + b \cdot \mathbb{1} \tag{1}$$

where $b \cdot \mathbb{1}$ denotes a bias term which takes the value $b \in \mathbb{R}$ and $\star$ denotes the convolution operation for cross correlation. Amounts to performing an exhaustive search of the pattern z over the image x further generate a score map f defined on a finite grid and match the maximum value in score or say response map regarded as the target location.

The model uses a discriminative approach, training the network on positive and negative pairs and adopting the logistic loss ℓ

$$\ell(y,v) = \log\left[1 + \exp(-y \cdot v)\right] \tag{2}$$

where v is the real-valued score of a single exemplar–candidate pair and $y \in \{-1, +1\}$ is its ground-truth label.

Training convolutional network φ with pairs that comprise an exemplar image and a larger search image will produce a score map $\mathcal{D}$. The loss of score map L is defined as the mean of individual losses

$$L(y,u) = \frac{1}{|\mathcal{D}|} \sum_{u \in \mathcal{D}} \ell(y[u], v[u]) \tag{3}$$

where $|\mathcal{D}|$ is the total number of elements in the score map, and u represents each position in the score map. The parameters of the convolutional network are obtained by applying stochastic gradient descent (SGD) to minimize loss L.

With time evolution, Li *et al.* proposed SiamRPN which is based on the Siamese framework incorporating region proposal networks (RPNs) for proposal generation. The correlation is computed in a group manner on both classification and regression branches as

$$f_\sigma(z,x)_{2k}^{\text{cls}} = \phi_\sigma(z) \star \phi_\sigma(x) \tag{4}$$

$$f_\varrho(z,x)_{4k}^{\text{reg}} = \psi_\varrho(z) \star \psi_\varrho(x) \tag{5}$$

where $\star$ denotes the convolution operation for cross correlation. Each point in $f_\sigma(z,x)_{2k}^{\text{cls}}$ contains a $2k$ channel vector, which represents for negative and positive activation of each anchor at the corresponding location on the original map. Similarly, each point in $f_\varrho(z,x)_{4k}^{\text{reg}}$ contains a $4k$ channel vector, which represents dx, dy, dw, and dh measuring the distance between the anchor and the corresponding ground truth. One year later, an advanced version named SiamRPN++ was proposed and achieved considerable advanced performance.

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B. Self-Attention

In the following subsections, we will propose the pertinent improvement models. For a clearer presentation, here we take the search branch $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$ as input for example.

1) Self-Channel Attention (SCA): To provide a clear description of the proposed SCA module, we first revisit the channel attention module in SENet (SE block). Given the input $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$, the squeeze step for the $\boldsymbol{c}$th channel can be formulated as follows:

$$z_c = \sum_{i=1}^{H}\sum_{j=1}^{W} \boldsymbol{x}_c(i,j)/(H\times W) \tag{6}$$

where $\boldsymbol{x}_c$ is the $\boldsymbol{c}$th monolayer feature of $\boldsymbol{X}$, and $\boldsymbol{z}_c$ is the same part of $\boldsymbol{Z}$: the output associated with the channel.

The following excitation step aims to fully capture channel-wise dependencies $\hat{\boldsymbol{X}}$, which can be formulated as:

$$\hat{\boldsymbol{X}} = \boldsymbol{X}\cdot\sigma(W_2\delta(W_1\boldsymbol{Z})) \tag{7}$$

where $\cdot$ refers to channel-wise multiplication, σ is the sigmoid function, δ is the ReLU function, and the sizes of W_1 and W_2 are set to $C*(C/r)$ and $(C/r)*C$, respectively, where r is a hyperparameter to vary the capacity and computation in the network; mostly setting $r=16$ achieves a good balance between accuracy and complexity.

Inspired by [35] and [36], the SCA module focuses on two aspects: 1) avoid dimension reduction to destroy the direct correspondence between the channel and its weight and 2) location information is introduced as an extra reference to enhance the channel identification capability of the module. The implementation details of the module are as follows.

Given the input $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$. First, we apply a pair of 2-D average pooling operations, their pooling kernels, respectively, are $(H,1)$ and $(1,W)$ to encode each channel along the horizontal coordinate and the vertical coordinate, respectively. The two corresponding output features are $\boldsymbol{PH} \in \mathbb{R}^{C\times H\times 1}$ and $\boldsymbol{PW} \in \mathbb{R}^{C\times 1\times W}$. The operational logic that gets the $\boldsymbol{c}$th channel of $\boldsymbol{PH}$ at height $\boldsymbol{h}$ can be formulated as

$$\boldsymbol{PH}_c^{\boldsymbol{h}} = \frac{1}{W}\sum_{i=1}^{W}\boldsymbol{X}_c(\boldsymbol{h},i) \tag{8}$$

where the subscript $\boldsymbol{c}$ indicates the $\boldsymbol{c}$th channel, and the superscript $\boldsymbol{h}$ represents the feature pixel of a specific height parameter $\boldsymbol{h}$. Similarly, the output of the $\boldsymbol{c}$th channel of $\boldsymbol{PW}$ at width $\boldsymbol{w}$ can be written as

$$\boldsymbol{PW}_c^{\boldsymbol{w}} = \frac{1}{H}\sum_{j=1}^{H}\boldsymbol{X}_c(j,\boldsymbol{w}). \tag{9}$$

Second, a kernel parameter K was calculated according to the channels number C and preset parameters α and β

$$t = \left[\log(C,\ 2)+\beta\right]/\alpha \tag{10}$$

$$K = \begin{cases} t, & \text{if } t \shortparallel 2 = 1 \\ t+1, & \text{else} \end{cases} \tag{11}$$

where $\shortparallel$ denotes take the remainder number operation. After obtaining K, build two 1-D convolutions, namely, conv1d_h and conv1d_w according to it, where the convolution kernel size is K and padding parameter is the integer of $(K/2)$ (padding ensure the convoluted result remains the same shape as input one). The input dimensions of these two convolutions are the height and width of the input feature $\boldsymbol{X}$, respectively, and the output dimensions are unity as 1.

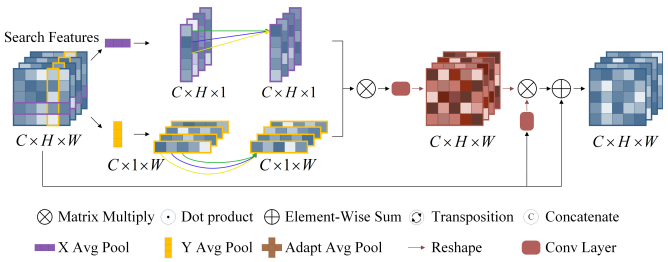


Fig. 3. Architecture of the self-channel attention module. As the first detailed structural diagram, the symbol corresponding meanings are involved at the bottom. Beware of the confusion of reshaping operations red connecting line and regular black line.

Third, feeding $\boldsymbol{PH}$ into conv1d_h and $\boldsymbol{PW}$ into conv1d_w captures local–interact channel interaction by considering every channel and its K neighbors and generates the channel weights' feature $\boldsymbol{NH} \in \mathbb{R}^{C\times 1}$ and $\boldsymbol{NW} \in \mathbb{R}^{1\times C}$. Combining $\boldsymbol{NH}$ and $\boldsymbol{NW}$ using matrix multiplication to get $\boldsymbol{N} \in \mathbb{R}^{C\times C}$.

Next, build 1×1 2-D convolution input with $\boldsymbol{X}$ and reshape output to get $\boldsymbol{V} \in \mathbb{R}^{C\times U}$ where $U = H\times W$. Combine $\boldsymbol{N}$, $\boldsymbol{V}$ and reshaped initial input $\boldsymbol{X}$ to get the final output feature map $\boldsymbol{Y}_{\mathbf{sca}} \in \mathbb{R}^{C\times H\times W}$ via residual connection operation as

$$\boldsymbol{Y}_{\mathbf{sca}} = \text{Re}(\lambda\cdot\boldsymbol{NV}+\boldsymbol{X}) \tag{12}$$

where the subscript **sca** of $\boldsymbol{Y}$ is the shortened form for SCA, Re is the shortened form of reshape operation, and λ is a learnable scalar parameter. Fig. 3 gives a schematic of the SCA module.

2) Self-Spatial Attention (SSA): Inspired by [36], [37], the SSA module divides the original feature into two 1-D coding processes, aggregate feature along with two spatial directions, respectively. Remote dependencies can be captured along one direction while precise location information can be retained from another direction. For each feature position, the pair-wise relationship between this position and all the positions are captured to mining global structure-related information. Convolutional layers with the translation-invariant property are used to efficiently exploit the global structural information.

Given the input $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$. First, we apply two separate 1×1 convolution layers on $\boldsymbol{X}$ to generate feature $\boldsymbol{Q}$ and $\boldsymbol{V} \in \mathbb{R}^{C'\times H\times W}$, separately, where $C' = (1/\sigma)\times C$, and σ is a preset parameter that controls the extent of channel reduction.

Second, two feature maps are then reshaped to $\bar{\boldsymbol{Q}}$ and $\bar{\boldsymbol{V}} \in \mathbb{R}^{C'\times N}$ where $N = H\times W$. Taking the feature vector at spatial position i in feature map $\boldsymbol{G} \in \mathbb{R}^{C'\times H\times W}$ as feature node $\boldsymbol{G}_i \in \mathbb{R}^{C'\times 1\times 1}$. The affinity correlation matrix $\boldsymbol{U} \in \mathbb{R}^{N\times N}$ of $\boldsymbol{X}$ can be generated via $\bar{\boldsymbol{Q}}$ and $\bar{\boldsymbol{V}}$

$$\bar{\boldsymbol{Q}} = \sum_{i=1}^{N}[\theta(\boldsymbol{X}_i)],\quad \bar{\boldsymbol{V}} = \sum_{i=1}^{N}[\phi(\boldsymbol{X}_j)] \tag{13}$$

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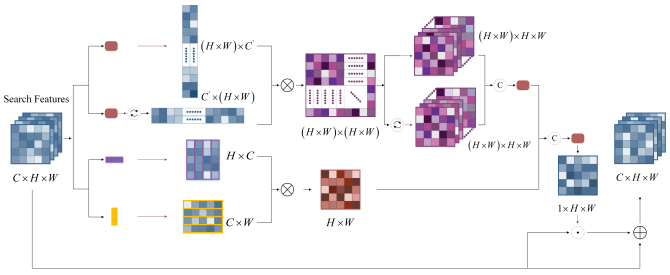


Fig. 4. Architecture of the self-spatial attention (SSA) module. The size of some of the feature maps is too large, and ellipsis is used to keep the specifications uniform. C' represents the channel numbers after dimension reduction.

$$\boldsymbol{U} = \bar{\boldsymbol{Q}}^T \bar{\boldsymbol{V}} = \sum_{i=1}^{N}\sum_{j=1}^{N}\left[\theta(\boldsymbol{X}_i)^T \cdot \theta\left(\boldsymbol{X}_j\right)\right] \tag{14}$$

where $\Sigma[\,]$ represents the repetitive combination operation, superscript T stands for the transpose, and θ and ϕ are the two embedding functions implemented by a 1×1 spatial convolutional layer followed by batch normalization (BN) and ReLU activation.

Third, through permute and reshape operations with input $\boldsymbol{U}$ get maps $\boldsymbol{GA}$ and $\boldsymbol{GB} \in \mathbb{R}^{N\times H\times W}$, concatenate them two in channel dimension, and through 1×1 convolutional layer get feature map $\boldsymbol{G} \in \mathbb{R}^{N\times H\times W}$. Fourth, sending $\boldsymbol{X}$ into a pair of 2-D average pooling operations, their pooling kernels are $(H, 1)$ and $(1, W)$, respectively, get feature maps $\boldsymbol{R}$ and $\boldsymbol{E}$. Combine them two through matrix multiplication to get $\boldsymbol{K} \in \mathbb{R}^{1\times H\times W}$.

Next, merge $\boldsymbol{K}$ and $\boldsymbol{G}$ in channel dimension and through translation invariant convolutional layers to get $\boldsymbol{W} \in \mathbb{R}^{1\times H\times W}$, and through the expand operation to the same size as $\boldsymbol{X}$ to get $\hat{\boldsymbol{W}}$. The final output feature map $\boldsymbol{Y}_{\mathbf{ssa}} \in \mathbb{R}^{C\times H\times W}$ are obtained as

$$\boldsymbol{Y}_{\mathbf{ssa}} = \mathrm{Re}\left(\varphi \cdot \hat{\boldsymbol{W}} \cdot \boldsymbol{X} + \boldsymbol{X}\right) \tag{15}$$

where the subscript **ssa** of $\boldsymbol{Y}$ is the shortened form for SSA, Re is the shortened form of reshape operation, and φ is a learnable scalar parameter. Fig. 4 gives a schematic of the SSA module.

C. Cross-Attention

Compared with self-attention, cross-attention [38]–[41] is still in a relatively preliminary stage of development. The core idea of cross-attention is to establish a correlation relationship between multiple input features via a crossover, and this relationship can not only realize feature fusion but also feedback to the inputs to realize the focus of interesting content. Such a property has mighty application potential for tasks with multiple original data input to the model, such as change detection and object tracking.

One major contribution of SiamRPN++ is to prove more extensive feature cross correlation or say fusion between two branches is conducive to the improvement of tracking performance. Served as two mainstays of the Siamese architecture, template and search branches, in earlier models [34], [42], [43] the feature from two branches are computed separately and only realize cross correlation in the last stage to make predictions.

Although SiamRPN++ does not use cross-attention in the proper sense, they are indeed highly similar in core idea, which indirectly proves the feasibility of cross-attention application in Siamese-based trackers. As expected, in the follow-up work SiamAttn [21], the authors designed a cross-attention module that provides a new feature crossover method and proved its effectiveness through ablation experiments. For clearer presentation, here we take template feature $\boldsymbol{Z} \in \mathbb{R}^{C\times h\times w}$ and search branch feature $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$ as the inputs of examples.

1) Cross-Channel Attention (CCA): Currently, there are relatively few models [40], [41] that construct independent cross-attention at the channel level. Among them, SiamAttn in the tracking field is more representative. The CCA modules in SiamAttn are divided into two submodules based on the different main branches. Taking the search branch as the main branch as an example, the authors first reshape the template feature $\boldsymbol{Z}$ to $\bar{\boldsymbol{Z}} \in \mathbb{R}^{C\times n}$ where $n = h \times w$. Then compute the cross-attention from the template branch by performing channel self-attention operations. Then the cross-attention computed from the template branch is encoded into the search feature $\boldsymbol{X}$. The whole process can be formulated as

$$\boldsymbol{A} = \phi_{\mathrm{row}}\left(\bar{\boldsymbol{Z}}\bar{\boldsymbol{Z}}^T\right) \in \mathbb{R}^{C\times C} \tag{16}$$

$$\bar{\boldsymbol{Y}} = \gamma \cdot \boldsymbol{A}\bar{\boldsymbol{X}} + \bar{\boldsymbol{X}} \in \mathbb{R}^{C\times N} \tag{17}$$

where ϕ_{row} is the softmax function implement row normalization, γ is a scalar parameter, and $\bar{\boldsymbol{X}} \in \mathbb{R}^{C\times N}$ where $N = H \times W$ is the reshape version of $\boldsymbol{X}$. The reshaped feature map $\boldsymbol{Y} \in \mathbb{R}^{C\times H\times W}$ is the output of submodule.

However, there are still two shortcomings in SiamAttn: 1) the setting that according to different main branches divided into two submodules, equivalent to twice the calculation process, which is relatively tedious and 2) in essence, the whole method is to obtain cross features by combining the other party with the mapping map from self-attention. It does not realize the actual direct crossover between two branch features. To overcome the above shortcomings, a concise and direct ameliorate module is proposed.

Input the template feature $\boldsymbol{Z} \in \mathbb{R}^{C\times h\times w}$ and search feature $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$, respectively. First, apply adaptive average pooling with preset parameter ε and reshape operations to get $\boldsymbol{J} \in \mathbb{R}^{C\times\varepsilon^2}$ and $\boldsymbol{P} \in \mathbb{R}^{\varepsilon^2\times C}$, and combine these two by matrix multiplication to get $\boldsymbol{K} \in \mathbb{R}^{C\times C}$. Second, define two 1×1 2-D convolution layers, and separately input $\boldsymbol{Z}$ and $\boldsymbol{X}$ get $\boldsymbol{Q} \in \mathbb{R}^{C\times n}$ and $\boldsymbol{V} \in \mathbb{R}^{C\times N}$ where $n = h \times w$ and $N = H \times W$. Combine $\boldsymbol{K}$, $\boldsymbol{Q}$, and $\bar{\boldsymbol{Z}} \in \mathbb{R}^{C\times n}$ with a residual connection and reshape operation to get $\boldsymbol{Y}_{\mathbf{cca}}(\boldsymbol{Z})$; in the same way get $\boldsymbol{Y}_{\mathbf{cca}}(\boldsymbol{X})$ by $\boldsymbol{K}^T$, $\boldsymbol{V}$, and $\bar{\boldsymbol{X}} \in \mathbb{R}^{C\times N}$. The process can be formulated as

$$\boldsymbol{Y}_{\mathbf{cca}}(\boldsymbol{Z}) = \mathrm{Re}\left(\rho \cdot \boldsymbol{K}\boldsymbol{Q} + \bar{\boldsymbol{Z}}\right) \tag{18}$$

$$\boldsymbol{Y}_{\mathbf{cca}}(\boldsymbol{X}) = \mathrm{Re}\left(\rho \cdot \boldsymbol{K}^T\boldsymbol{V} + \bar{\boldsymbol{X}}\right) \tag{19}$$

where the subscript **cca** of $\boldsymbol{Y}$ is the shortened form for CCA, Re is the shortened form of reshaping operation, ρ is a learnable scalar parameter, and the superscript T stands for the transpose. Fig. 5 gives a schematic of the CCA module.

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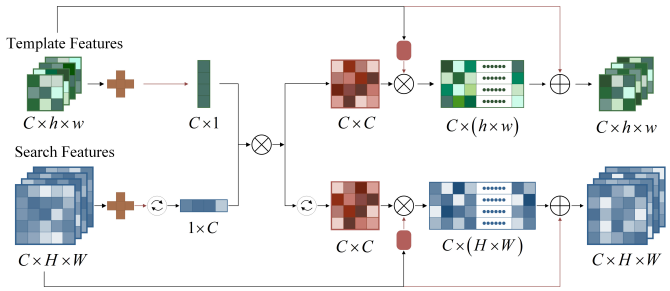


Fig. 5. Architecture of the CCA module. The yellow cross represents adaptive average pooling; for clarity, we use global pooling with 1×1 output. Ellipsis is also used to control the image size.

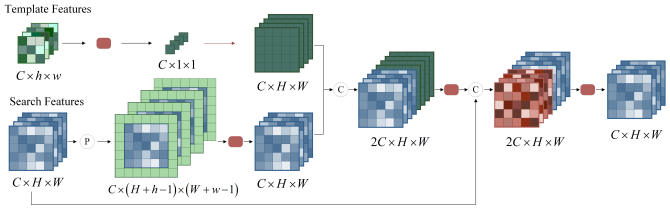


Fig. 6. Architecture of the cross-spatial attention module. As seen, the module has two branches' inputs but only outputs one search feature map. This design aims to avoid the chain failure reaction caused by the wrong spatial information contamination template.

2) Cross-Spatial Attention (CSA): Up to now, there are rarely studies on cross-spatial attention (CSA), especially blank in the field of tracking. Even in the relatively comprehensive study article SiamAttn, the proposed so-called cross-attention mechanism still only involves the channel level, and there is no mention of how to achieve spatial level cross.

The apparent reason is not hard to figure out, the implementation of crossover requires the same size attention map, and this is easy to achieve for SCA, SSA, and CCA modules. But for spatial attention, the dimensions of "spatial attention map" for template and search image are $\mathbb{R}^{n\times n}$ and $\mathbb{R}^{N\times N}$, respectively, where $n = h \times w$ and $N = H \times W$. Obvious of these two scale mismatches, unable simple mechanized, apply the previous pattern.

However, there might be another underlying reason besides the above obvious reason: The idea of CSA has many similarities with the cross correlation mechanism of Siamese-based trackers. In the case of design difficulties due to size mismatch, the necessity of spatial crossover is indeed controversial.

Even so, the rationality of CSA is still worth believing. The "multilayer-wise aggregation" and "multilevel cascade RPN" strategies adopted in SiamRPN++ effectively improve the algorithm performance. Their essence can be summarized as the core idea of "multivariate information joint decision." In contrast, in the early tracking models [34], [42], [43], the way of "simplex feature extraction" and "single-layer correlation determined location" has fallen behind. The CSA enhances not only the feature diversity extent of the model but also an insurance for follow-up correlation map decisions, effectively relieving the decision-making pressure.

The key to CSA implementation is how to efficiently concatenate different-sized feature maps (template and search). Generally, there are two conventional approaches.

1) Cross correlation [34], [44], [45], essentially is a standard convolution, template feature as the convolution kernel to carry out convolution instance feature, and the convolution output is the similarity fraction of each position. But its downside is it is a handcrafted parameterless module that cannot fully benefit from large-scale learning off-line data.
2) Slide concatenation, a naive strategy to perform concatenation on different-sized feature maps. First split the search region feature map into multiple subwindows with the same size as the template feature, then different subwindows and the template are concatenated along the channel axis, and finally, each spliced feature is output through a convolutional fused feature. But this approach requires many times to repeat the sliding window and splicing operations, although off-line learning can be achieved but with a vast calculating amount.

In contrast, asymmetric convolution (AC) [46] replaces this direct convolution on the concatenated feature map with two independent convolutions: First, it separately convolves the two feature maps using kernels of the same size of the template feature map. Then, it computes the summation on these different-sized feature maps through broadcasting. This could circumvent the disadvantages of the above two while realizing mathematically equivalent procedure. Precisely based on this idea, the CSA module is proposed.

Given inputs $\boldsymbol{Z} \in \mathbb{R}^{C\times h\times w}$ and $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$. First, calculate extension parameters according to h and w, and pad $\boldsymbol{X}$ by them to get $\boldsymbol{X}_{\mathbf{pad}} \in \mathbb{R}^{C\times(H+h-1)\times(W+w-1)}$. Second, define two $h \times w$ convolutional layers, input $\boldsymbol{X}_{\mathbf{pad}}$ into one layer to get $\boldsymbol{V} \in \mathbb{R}^{C\times H\times W}$, $\boldsymbol{Z}$ into another layer, and then combine expand operation to get $\boldsymbol{Q} \in \mathbb{R}^{C\times H\times W}$. The process can be formulated as

$$\boldsymbol{V} = \phi_{\langle(c\to c)|(h,w)\rangle}\left(\boldsymbol{X}_{\mathbf{pad}}\right) \tag{20}$$

$$\boldsymbol{Q} = \left[\psi_{\langle(c\to c)|(h,w)\rangle}(\boldsymbol{Z})\right]_{\exp(\boldsymbol{X})} \tag{21}$$

where ϕ and ψ represent the convolution operations, subscript $\langle(c1 \to c2)|(h1, w1)\rangle$ represents the convolution parameters: input channels $c1$ and output channels $c2$, and the convolution kernel size is $(h1, w1)$. $[\,]_{\exp(X)}$ is the expand operation with $\boldsymbol{X}$ size output.

Third, define two 1×1 convolutional layers, concatenate $\boldsymbol{V}$ and $\boldsymbol{Q}$ at channel level, send concatenated feature in one layer to get $\boldsymbol{S} \in \mathbb{R}^{C\times H\times W}$, concatenate $\boldsymbol{X}$ and $\boldsymbol{S}$ in the same way, and send into another layer get output $\boldsymbol{Y}_{\mathbf{csa}} \in \mathbb{R}^{C\times H\times W}$. The process can be formulated as

$$\boldsymbol{S} = \beta_{\langle(2c\to c)|(1,1)\rangle}(\boldsymbol{V} \oplus \boldsymbol{Q}) \tag{22}$$

$$\boldsymbol{Y}_{\mathbf{csa}} = \xi_{\langle(2c\to c)|(1,1)\rangle}(\boldsymbol{X} \oplus \boldsymbol{S}) \tag{23}$$

where the subscript **csa** of $\boldsymbol{Y}$ is the shortened form for CSA. β and ξ are the convolution operations. $\oplus$ represents the concatenation of channel dimensions. Fig. 6 gives a schematic of the CSA module.

D. Feature Combination

Extracted feature combination of the template and search images generating the corresponding similarity feature has

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become a common strategy for subsequent object localization. Work [21] performing cross-correlation on multiscale features is the most common pattern. Besides, scholars [47] have also achieved good performance based on a unique single-scale way.

However, neither approach fits well with our work, considering the JSANet obtains several feature maps of the template and search images during the feature extraction stage, facing many such feature maps; if all of them adopt multiscale correlation, it would bring huge computational overhead. In addition, if fusing them into one map and then conducting the newer single scale fusion strategy, considering the different obtained sources of these maps, their relative importance also varies, and simply using point-wise summation seems a bit rough.

Viewing the above problems, the feature fusion strategy is divided into two portions.

First, dynamic learning weights' adaptive is determined for the participation degree of each module; it is further multiplied with obtained attention-wise feature maps and then combine with an element-wise sum to generate the fusion-attentional feature

$$\boldsymbol{Y}_{\boldsymbol{fa}} = \begin{bmatrix} \lambda \\ \mu \\ \nu \\ \xi \end{bmatrix} \begin{bmatrix} \boldsymbol{Y}_{\mathbf{sca}} & \boldsymbol{Y}_{\mathbf{ssa}} & \boldsymbol{Y}_{\mathbf{cca}} & \boldsymbol{Y}_{\mathbf{csa}} \end{bmatrix} \tag{24}$$

where the subscript **fa** of $\boldsymbol{Y}$ is the shortened form for fusion-attentional, and subscripts **sca**, **ssa**, **cca**, and **csa** are shortened forms for SCA, SSA, CCA, and CSA, respectively. λ, μ, ν, and ξ are the four learnable scalar parameters corresponding to the four attention module outputs, respectively. Their values can be calculated by softmax function with δ_{λ_c}, δ_{μ_c}, δ_{ν_c}, and δ_{ξ_c} as control parameters, respectively

$$\lambda_c = \frac{e^{\delta_{\lambda_c}}}{e^{\delta_{\lambda_c}} + e^{\delta_{\mu_c}} + e^{\delta_{\nu_c}} + e^{\delta_{\xi_c}}}. \tag{25}$$

We use global average pooling layers and full connection to compute the weight scalar maps, δ_λ, δ_μ, δ_ν, and δ_ξ from $\boldsymbol{Y}_{\mathbf{sca}}$, $\boldsymbol{Y}_{\mathbf{ssa}}$, $\boldsymbol{Y}_{\mathbf{cca}}$, and $\boldsymbol{Y}_{\mathbf{csa}}$, respectively. Then they can be learned through standard back-propagation.

Second, three levels (conv3/4/5) of feature outputs are fed into three SiamRPN to collaboratively infer the target localization. For each level, the features of all the other levels are resized to the same shape and spatially fused according to the learned weight maps

$$\boldsymbol{M}^l = \alpha^l \boldsymbol{M}^{3\to l} + \beta^l \boldsymbol{M}^{4\to l} + \gamma^l \boldsymbol{M}^{5\to l} \tag{26}$$

$$\boldsymbol{A}^{\text{cls}}_{w\times h\times 2k} = \sum_{l=3}^{5} \left[\boldsymbol{X}^l_{\text{cls}} \star \boldsymbol{Z}^l_{\text{cls}} \right] \tag{27}$$

$$\boldsymbol{A}^{\text{reg}}_{w\times h\times 4k} = \sum_{l=3}^{5} \left[\boldsymbol{X}^l_{\text{reg}} \star \boldsymbol{Z}^l_{\text{reg}} \right] \tag{28}$$

where α^l, β^l, and γ^l refer to the importance weights for the feature maps at three different levels to level $l \in \{3, 4, 5\}$, and $\boldsymbol{M}^{n\to l}$ denotes the feature maps $\boldsymbol{M}$ resized from level n to level l. $\star$ denotes the convolution operation, and $\boldsymbol{A}^{\text{cls}}_{w\times h\times 2k}$ represents for negative and positive activation of each anchor at a corresponding location on the original map; similarly, $\boldsymbol{A}^{\text{reg}}_{w\times h\times 4k}$ represents dx, dy, dw, and dh measuring the distance between the anchor and the corresponding ground truth. α^l, β^l, and γ^l get their values in the same way that λ, μ, ν, and ξ do.

IV. Experiment and Analysis

This section is divided into five sections. First, by comparing with classical datasets, we show the specifications, advantages, and disadvantages of Jilin-1 and ISS. Next, the training and testing stage details such as experimental platform, design parameters, data scale, and comparison methods are introduced, to have a specific understanding of the whole implementation process. Following, qualitative and quantitative experiments' omnidirectional contrast the JSANet with 15 state-of-art trackers in terms of visualizations, plots, and tables, respectively. Finally, an ablation analysis study is provided to verify the effects of each proposed component.

A. Experimental Datasets Description

In this article, two satellite video datasets, respectively, named Jilin-1 and ISS were used to verify the effectiveness of the proposed JSANet. The Jilin-1 dataset contains nine video sequences acquired from Jilin-1 satellites, purchased from Changchun-based ChangGuang Satellite Technology. The sequences contain 320 frames with 10 fps and 0.92–1.16-m spatial resolution, a total including 461 targeted vehicles. The ISS dataset has one video sequence acquired from the International Space Station (ISS) provided by Deimos Imaging and UrtheCast (Vancouver, BC, Canada) [48], containing 1032 frames with 30 fps and 1-m spatial resolution, including 61 goals. Since there is only one video inside, we used only ISS data for the test, to avoid inflated results caused by participating in training and testing at the same time.

To get a clearer description of Jilin-1 and ISS, we use the benchmark dataset in wide area motion imagery (WAMI) research Columbus large image format (CLIF) [49] to make a comparison and show their characteristics. Starting with the advantage part, the Jilin-1 and ISS datasets hold three advantages over CLIF as follows.

1) *Richer Coloring Information:* The data format of CLIF is a single-channel grayscale image, while the Jilin-1 and ISS datasets could provide optical remote sensing video data as rich coloring information as conventional natural data. This is not only more visually friendly but also to some extent more convenient for the transformation and utilization of classical natural algorithms. The richer color feature also provides more possibilities for the application of deep learning algorithms.
2) *Wider Range of Scanning (Imagery):* The imagery platform of the CLIF dataset is aircraft, while the Jilin-1 and ISS datasets are the satellite and space station in solar synchronous orbit, and there is a huge distinction in the photograph height between them. Profiting from higher imagery height, Jilin-1 and ISS can scan over a wider range. In addition, remote sensing satellites are safer

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TABLE I
COMPARISON OF PRIMARY ATTRIBUTES BETWEEN CLIF, JILIN-1, AND ISS DATASETS

Dataset	Photograph Height	Coverage Area	Frame Rate	Data Format	Resolution	Per Vehicle Size
CLIF	$4-8$ km	5×5 km^2	$1-2$ fps	Grayscale	$0.2-0.5$ m	20×30 px
Jilin-1	500 km	11×4.6 km^2	10 fps	RGB	$0.92-1.16$ m	6×10 px
ISS	400 km	3.8×2.1 km^2	30 fps	RGB	1 m	6×10 px

and more convenient than aircraft for the monitoring of restricted areas.

3) *Higher Frame Rate:* Compared with the frame rate of 2 fps in the CLIF dataset, Jilin-1 and ISS reach a higher frame rate. More sufficient temporal information is helpful to record the changing situation of the object, especially since it is effective for tracking tasks to narrow down the search range, and the consistency of the tracking process is largely guaranteed.

However, the disadvantages of Jilin-1 and ISS are equally obvious; the higher imagery height imagery caused the common occurrence of data quality inferiority and object feature scarcity. Although Jilin-1 and ISS have a relatively excellent resolution ratio and rich coloring information, the situation for vehicle objects remains grim. As seen from Table I, in the Jilin-1 and ISS datasets, the average pixel number occupied by a vehicle is only 10%–25% of one in CLIF. In addition, due to the high photograph distance, atmospheric particles, clouds, and other factors will harm the imaging quality, further blur the boundary between the object and the background (also result in definition, brightness, and contrast varying between frames), and increase the difficulty of tracking. The example of Google Earth, WPAFB, CLIF, Jilin-1, and ISS frame with ground-truth annotations is shown in Fig. 7.

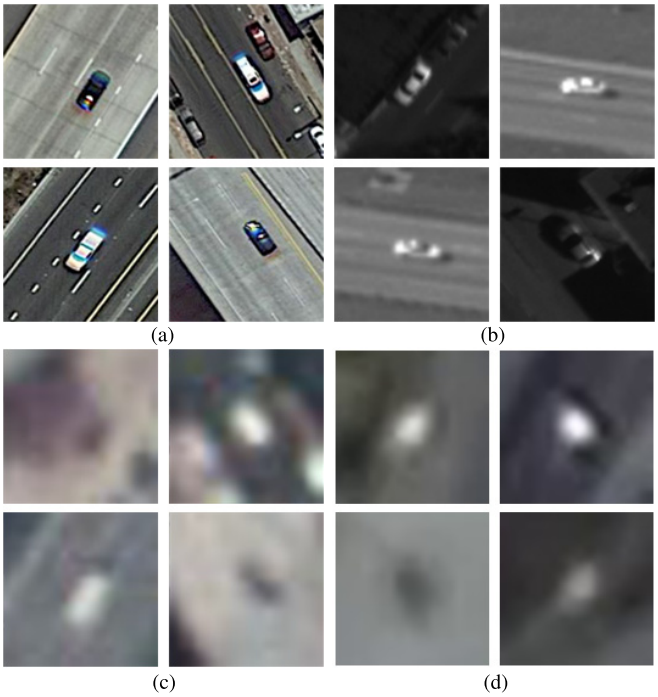


Fig. 7. Compared with previous datasets, the vehicles in Jilin-1 and ISS lack appearance characteristics, and the boundary between the object and the surrounding environment is blurred, which increases the difficulty of tracking, especially when the surrounding environment is complex or the image quality of the interframe changes significantly. (a) Google Earth. (b) WPAFB/CLIF. (c) Jilin-1. (d) ISS.

B. Implementation Details

1) Training: We use ResNet-50 pretrained on ImageNet [50] as the backbone network. The JSANet is implemented using the PySOT platform with Python 3.6 and PyTorch 1.3.0, synchronized SGD on 2 NVIDIA Tesla V100 GPUs, with each GPU hosting 48 images, and hence, the minibatch size is 96 images per iteration. There are 20 epochs in total, and each epoch uses 10^6 training pairs. For the first 10 epochs, we start with a warmup learning rate of 10^{-3} to train the joint attention networks, while freezing the parameters of the backbone. For the remaining epochs, the backbone network is unfrozen, and the whole network is trained end-to-end with a learning rate exponentially decayed from 5×10^{-3} to 5×10^{-4}. The weight decay and momentum are set to 10^{-5} and 0.9, respectively.

In addition to the basic parameters to run the program, there are four preset hyperparameters in our method, respectively, which are α and β in self-channel, σ in self-spatial, and ε in cross-channel. After a series of comparative tests, we finally locked these parameters at $\alpha = 2$, $\beta = 1$, $\sigma = 16$, and $\varepsilon = 7$.

In terms of training data, four Jilin-1 video sequences of Beijing (China), Hong Kong (China), Bogota (Columbia), and San Diego (USA), including 213 vehicle targets, were used for training. Besides, except for the basic Jilin-1 data, we also used part of natural tracking datasets [51]–[55] and mixed them in a 50–50 split ratio. The reason for doing this is that the data scale of Jilin-1 is small, and the data expansion achieved by mixed is more conducive to depth training, which is conducive to improving the accuracy and robustness of the model.

2) Testing: We choose to use KCF [56], C-COT [57], ECO [58], RT-MDNet [59], SiamDW [60], SiamRPN++ [45], SuperDiMP, SiamCAR [61], SiamBAN [62], KYS [63], PrDiMP [64], TrDiMP [65], TrTr [66], Nonlocal [67], and Triplet [68], a total of 15 models to compare with the JSANet algorithm. Among them, RT-MDNet is the classic tracking algorithm based on CNN feature representation. SiamRPN++ is the baseline of the JSANet; SiamDW, SiamCAR, and SiamBAN are the Siamese architecture methods. KCF, C-COT, ECO, SuperDiMP, and PrDiMP TrDiMP are the discriminative trackers series. TrDiMP and TrTr are pioneering works that applied transformer in the tracking field. Nonlocal and

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Triplet as classical self-attention methods are embedded in SiamRPN++ architecture to compare with our method.

As for the test datasets, a total of 248 targets in five Jilin-1 scenes of San Francisco (USA), Derna (Libya), Adana (Turkey), Minneapolis (USA), and Muharraq (Bahrain) were used for testing. In addition, all 61 targets in the ISS dataset were used for testing. These experiments are conducted on a server with 1 NVIDIA Tesla V100 GPU. In terms of evaluation metrics, we use the common one-pass evaluation method mentioned in OTB to evaluate the performance of the proposed tracker JSANet and other related contrast trackers. The quantitative evaluation uses two evaluation metrics: precision and success rate calculation.

C. Qualitative Experiment

As mentioned above, the test datasets consist of six scenes with a total of 309 targets. To comprehensively evaluate the JSANet, we screened out some targets and divided them into two experimental scenes including similar interference (SI) and BC according to their characteristics, and then visualized the tracking effects of each algorithm upon them.

To avoid the impact of too many object boxes' stacking, seven representative algorithms were selected for display. Use different colors to represent these algorithms, and the corresponding relationship is shown at the bottom of Fig. 8. Corresponding to the above-mentioned two special scenes, two visual figures are shown. Each figure contains six targets, and each target is divided into upper and lower images. The upper image only shows the ground truth and the result of our method, and the lower image contains eight methods including the JSANet and seven comparison algorithms.

1) Similar Interference: A total of 101 targets were screened out to match this situation, and the visualization results of samples are shown in Fig. 8. The targets belonging to SI are mainly divided into two types. The first type is that the appearance of the target is a relatively empty blur [e.g., 8(a) and 8(e)] due to various reasons. The second type is the color of target highly similar [e.g., 8(b) and 8(f)] to the environment, such as white vehicles on cement roads and black vehicles on asphalt roads.

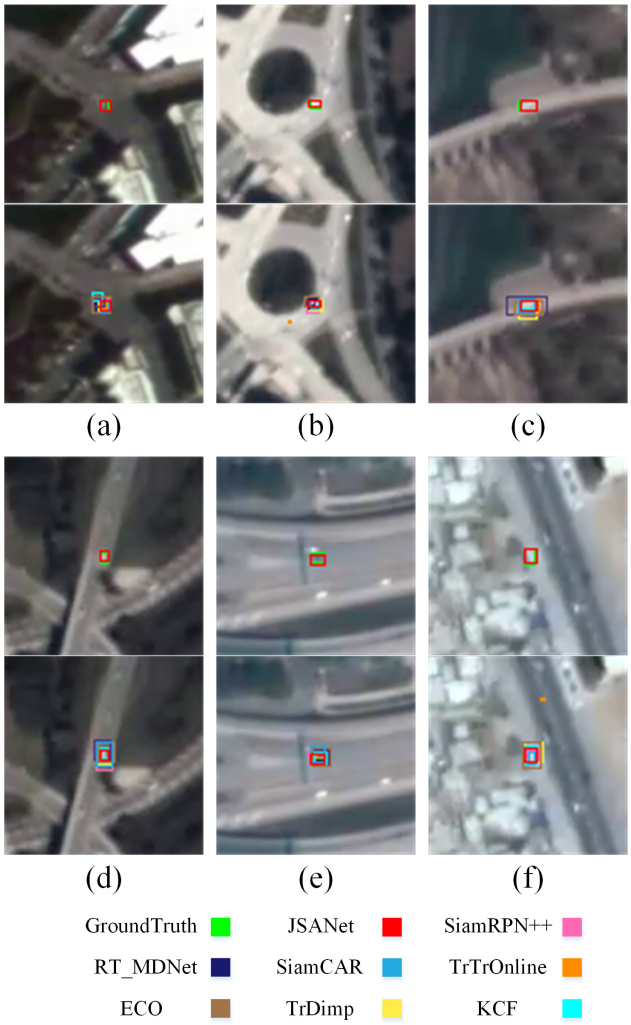


Fig. 8. Experimental results' visualization for similar interference (SI). The samples of (a), (d), and (e) belong to the emptiness blur situation. Although the color of the target is not close to the color of the road surface, it leads to tracking drift due to their fuzzy appearance. The samples of (b), (c), and (f) belonging to their color confuse with backgrounds further leading to tracking difficulties. Each algorithm uses the corresponding color to mark the target.

Poor discrimination between the target and the background greatly affected the discrimination ability of trackers. The most unsatisfactory situation (such as RT-MDNet and SiamCAR) is the missing target in the first few frames and the bounding box jitter at the starting point. The better situation (KCF and TrDiMP) is tracking drift: could barely follow the target, but the scale of the bounding box is highly unstable, and exhibited peak confidence also has an obvious error with real target positions. Even if there is force constraint on the range of the bounding box, the improved effect is also very limited. In contrast, the proposed attention mechanism could capture the subtle differences between the target and the background, ensuring the rationality of bounding boxes while accurately tracking the target continuously.

2) Background Changes: After screening, a total of 78 targets belong to this category, and the visualization results of samples are shown in Fig. 9. The scenes belonging to BCs can be subdivided into environmental changes and road surface changes. Environmental change refers to the trees and shadows appearing near the target causing the "approximate occlusion" effect. Road surface changes are caused by road material changes (e.g., Fig. 9(d) cement and asphalt switching) or halfway mark lines' confusion [e.g., Fig. 9(a) and (e)]. Among them, the formation conditions of environmental changes are harsh, and the actual occurrence frequency is less. In contrast, the occurrence of road surface changes, especially the marker lines' confusion is very common.

When the environment and road surface meantime become consistent with the color of the target, the majority of trackers (such as TrTrOnline and TrDiMP) will drift at the junction further losing the target. However, the confusion caused by the marker line will not cause target missed, but greatly affect target identification, leading to exhibited peak confidence offset and target scale change. At this time, if image quality

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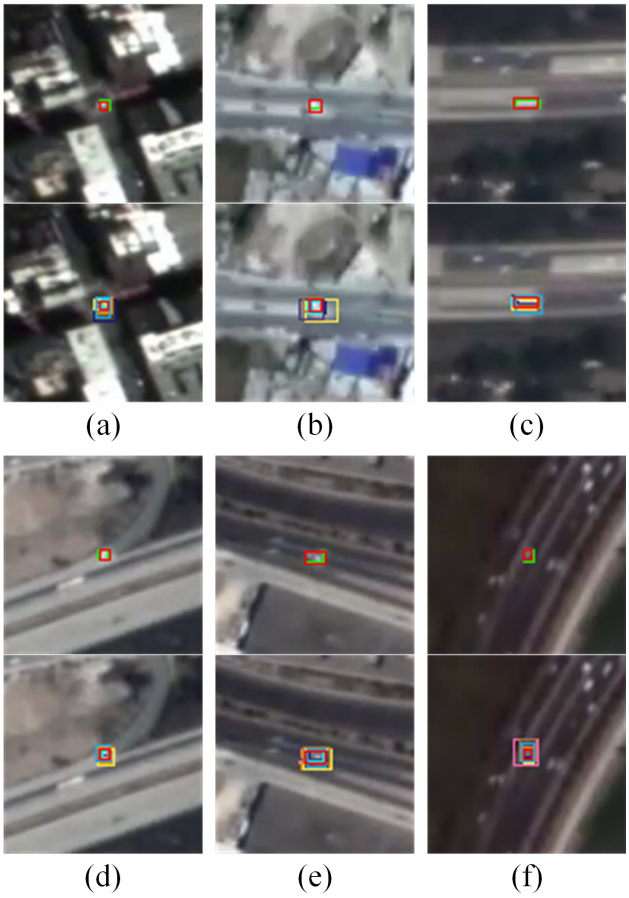


Fig. 9. Experimental results' visualization for background changes (BCs). The samples of (a), (e), and (f) due to halfway mark lines' confusion cause bounding box regression chaos. The samples of (b), (c) and (d) belong to road surface changes, easy to cause the trackers to stay at transform junction.

fluctuation or similar targets appear around, it is easy to cause target missed or error tracking. In this article, the target feature representation model constructed using the attention mechanism effectively strengthened the description of the target and improved the contrast between the target and the background.

D. Quantitative Experiment

In this section, we use the one-pass evaluation, which is a common evaluation method used in OTB, to evaluate the performance of the proposed tracker JSANet and other related trackers. The quantitative evaluation uses two criteria: precision and success rate.

1) Overall Performance: Fig. 10 shows the comparison results of 15 state-of-the-art algorithms and JSANet on the Jilin-1 test dataset. The 2nd and 3rd columns of Table II are the result of ISS. Notably, the approach proposed in this article (JSANet) shows the top performance according to both success rate and precision. Compared with the backbone model SiamRPN++, JSANet has improvements of 2.11% on precision and 2.39% on success in Jilin-1 data, 7.40% on precision, and 3.91% on success in ISS data, respectively.

As seen from the precision plot of Jilin-1, the curve rising trend of the top ten methods is relatively consistent. Even the existing reversal phenomenon (such as RT-MDNet and C-COT) does not draw a significant gap. In contrast, methods in success plot exhibit obvious differences, especially in the interval from 0.4 to 0.8 of overlap threshold, and the curves of C-COT and ECO show cliff fall, which is in strongly contrast to their top ranking in the precision plot. This phenomenon shows that although these methods can barely follow the target, they cannot correctly judge the boundary of the target. They are not tracking the target so much as tracking feature changes. In contrast, the proposed attention mechanism could more effectively capture the details of weak feature satellite targets while removing peripheral interference noise, achieving more accurate and efficient tracking performance.

2) Similar Interference: To further evaluate JSANet, it was compared with other algorithms under different experimental conditions. Fig. 11 shows the results of the experiment with SI, generally shortened to SI. The 4th and 5th columns of Table II are the SI result of ISS. Among all the tracking algorithms, JSANet ranks first for precision 69.18% and a success rate of 40.02% in Jilin-1 data, compared with SiamRPN++, which has significant improvements of 9.09% on precision and 5.73% on success, respectively.

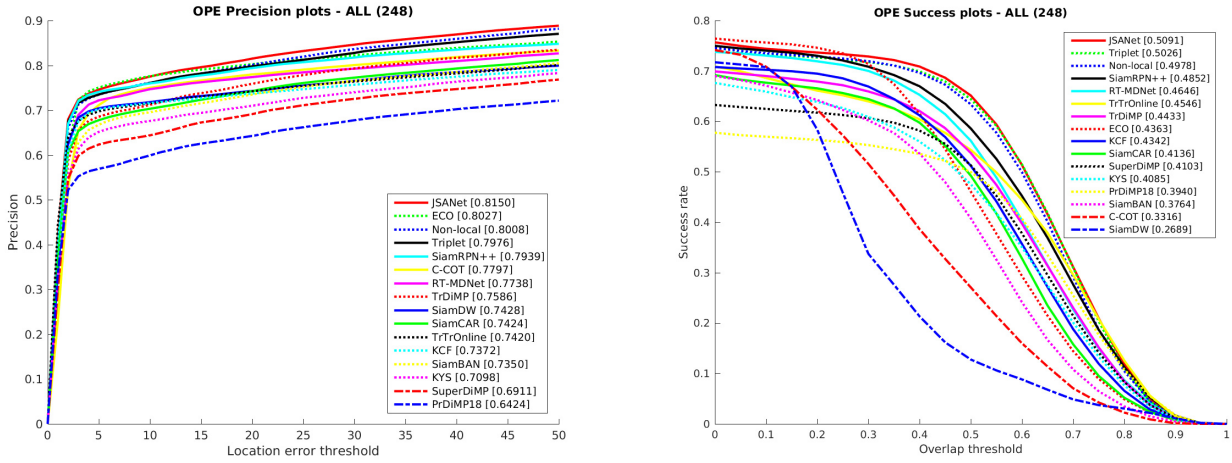
As seen from the precision plot, JSANet and top three comparison methods keep the same order. But from the view of precision decline, compared with the overall results, the decrease of JSANet is minimal, indicating that our method has better relative stability. The large precision decrease of other methods indicates that they lose targets more frequently in face of interference scenes. In the success plot, JSANet stays ahead in the entire threshold and the curve fell more smoothly. It shows that the proposed attention method effectively enhances the feature representation of the target and improves the contrast between the target when faced with approximate feature scenes.

3) Background Changes: Fig. 12 shows the results for the changes in backgrounds. Among all the experimental tracking algorithms, JSANet ranks first in the precision of 80.78% and a success rate of 48.00%, compared with SiamRPN++, which has significant improvements of 6.31% on precision and 5.03% on success, respectively.

By comparing the overall results, it can be seen that the accuracies of algorithms do not decrease so much. This is because in a video sequence, the appearance frames of BCs often only account for only part of global, before and after changes, and the algorithms can often maintain good results. As seen, in the success plot, the performance of C-COT and ECO is not ideal, which again shows that the correlation filter lacks the accurately predicting weak feature targets' ability in complex scenes. The reason for this situation is that, on one hand, lacking regression strategies such as RPN, on the other hand, their feature extraction pattern is not suitable for remote sensing small targets.

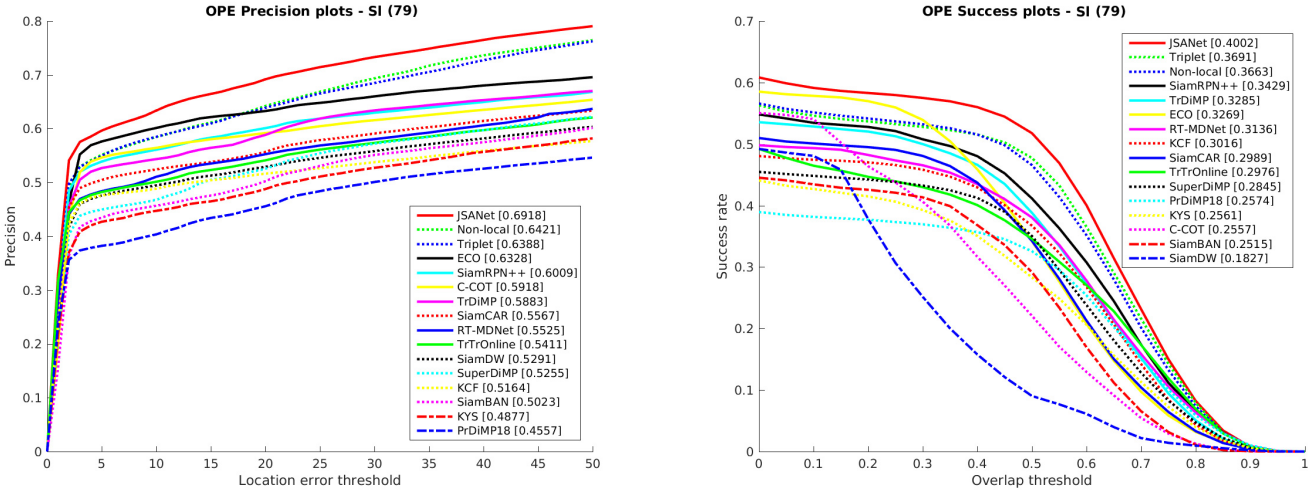
Another example of feature extraction pattern negative effecting performance was SiamDW-fc, which is the bottom of three success plots. Because its receptive field is too large, the feature of the target is directly covered by the surrounding background. Therefore, it is necessary to design a targeted architecture for weak feature targets.

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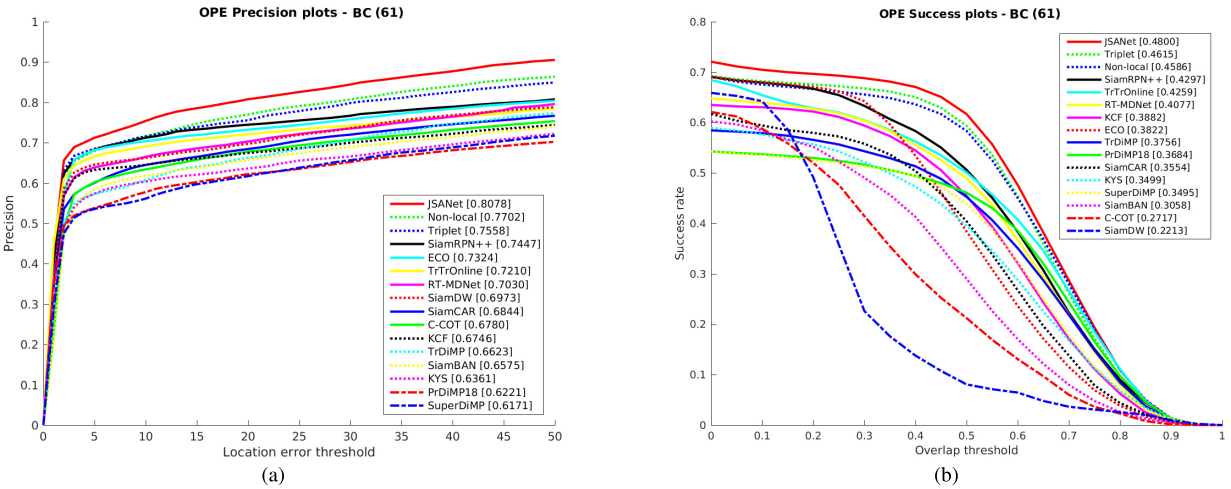
(a) (b)

Fig. 10. (a) Precision and (b) success rate of the comparisons with 15 state-of-the-art methods for overall performance.



(a) (b)

Fig. 11. (a) Precision and (b) success rate of the comparisons with 15 state-of-the-art methods for SI.



(a) (b)

Fig. 12. (a) Precision and (b) success rate of the comparisons with 15 state-of-the-art methods for BCs.

E. Effect Analysis of Preset Parameters

As mentioned before, there are four preset hyperparameters in our method, respectively, which are α and β in self-channel, σ in self-spatial, and ε in cross-channel. They were divided into three groups according to different modules. The experiments were carried out in one group, while the other two

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TABLE II
TRACKING RESULTS OF DIFFERENT METHODS IN THE ISS DATASET

	All (61)		SI (22)		BC (17)	
Method	Pre	Suc	Pre	Suc	Pre	Suc
JSANet	**0.8507**	**0.5524**	**0.6355**	**0.3598**	**0.7966**	**0.4995**
Triplet	0.8253	0.5394	0.6277	0.3298	0.7886	0.4885
Non-local	0.8192	0.5339	0.6105	0.3254	0.7842	0.4869
SiamRPN++	0.7767	0.5133	0.5971	0.3035	0.7531	0.4418
TrTrOnline	0.8157	0.5163	0.5634	0.3090	0.7802	0.4472
RT_MDNet	0.8081	0.4718	0.5742	0.2878	0.7294	0.4410
TrDiMP	0.7680	0.4298	0.5807	0.2753	0.7014	0.4182
ECO	0.7398	0.4946	0.5907	0.3081	0.7776	0.4305
KCF	0.7283	0.4468	0.5393	0.2903	0.7061	0.4279
SuperDiMP	0.7268	0.3584	0.5590	0.2579	0.6023	0.3848
SiamCAR	0.6866	0.4725	0.5601	0.2663	0.7453	0.4052
KYS	0.6755	0.3991	0.4705	0.2481	0.6638	0.3806
PrDiMP	0.5511	0.4216	0.5005	0.2357	0.6871	0.3871
SiamBAN	0.4910	0.4249	0.5205	0.2155	0.6965	0.3730
C-COT	0.4627	0.4454	0.5832	0.2563	0.7053	0.3219
SiamDW	0.4307	0.4869	0.5499	0.1784	0.7270	0.2941

TABLE III
RESULTS WITH VARIOUS NUMBERS OF PRESET PARAMETERS IN THE JILIN-1 AND ISS DATASETS

			Jilin-1		ISS		
α	β	K	Pre	Suc	Pre	Suc	FPS
3	0	[3, 3, 3]	0.8022	0.4999	0.8426	0.5495	**33.82**
2	0	[5, 5, 5]	0.8061	0.5089	0.8434	0.5488	33.16
2	1	[5, 5, 7]	**0.8150**	0.5091	**0.8507**	**0.5524**	33.10
2	3	[7, 7, 7]	0.8112	**0.5103**	0.8491	0.5516	32.97
-	-	[3, 5, 9]	0.8128	0.5096	0.8473	0.5459	32.65
1	0	[9, 11, 11]	0.8070	0.5086	0.8455	0.5457	32.12
-	-	[3, 5, 11]	0.8112	0.4989	0.8479	0.5504	32.08
σ			Pre	Suc	Pre	Suc	FPS
32			0.7986	0.4984	0.8388	0.5421	**35.47**
16			0.8150	0.5091	0.8507	0.5524	33.10
8			0.8159	0.5074	0.8511	0.5497	30.81
4			**0.8207**	**0.5103**	0.8587	**0.5541**	29.57
2			0.8173	0.5041	**0.8589**	0.5538	28.99
ε			Pre	Suc	Pre	Suc	FPS
15			0.8135	**0.5111**	**0.8541**	0.5513	29.11
7			**0.8150**	0.5091	0.8507	**0.5524**	33.10
3			0.7994	0.4995	0.8491	0.5487	34.27
1			0.7983	0.4983	0.8303	0.5421	**34.95**

groups were kept as preset fixed parameters through a series of alternate experiments, and the results of effect analysis are consolidated in Table III.

1) α and β: The purpose of setting α and β is to determine the parameter K of 1-D convolution, which determines the range and convergence of local interaction. Analysis of the experimental results shows that compared with the fixed K value, feature maps with different channel numbers often have better performance when selecting different K values according to their own volumes. In this case, the deeper ones, the ones with more channels, tend to have a slightly larger K value. This may be because the deeper and more abstract the information, the more likely it is to exploit hidden dependencies using long-range interactions. Although the optimal K value of the model can be achieved by fine-tuning, it is a waste of computing resources and time. In contrast, the combination of exponential function and adjustable parameter can obtain a fine trade-off of the fixed parameter under the condition of controllable cost and has better universality.

2) σ and ε: These two parameters of different modules are discussed together because they both directly determine the model size and computation burden. Their primary task is to investigate the trade-off between performance and computational cost mediated by these hyperparameters. The results show that the performance of the model does not increase monotonically with the continuous increase in the number of parameters. It is not suitable for them to fix the parameters, and some attempts should be made to select the best parameters. Fortunately, their optional parameters are often regular, and the search process of parameters is not too tedious. In addition, the FPS of the model often plays a key role in the selection of parameters. Especially for a tracing task like ours, the model must run more efficiently than the frame rate of the data, and preferably with some redundancy to ensure stable real-time tracking.

After the analysis of the test results, we finally selected these parameters as $\alpha = 2$, $\beta = 1$, $\sigma = 16$, and $\varepsilon = 7$. These four parameters no longer change in other experiments except in this section.

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TABLE IV
PERFORMANCE OF THE ABLATION EXPERIMENT IN THE JILIN-1 AND ISS DATASETS

	All				SI				BC			
					Jilin-1							
Method	Pre	Δ	Suc	Δ	Pre	Δ	Suc	Δ	Pre	Δ	Suc	Δ
Baseline + SA	0.8075	1.36%	0.5053	2.01%	0.6656	6.47%	0.3759	3.30%	0.7802	3.55%	0.4659	3.62%
Baseline + CA	**0.8123**	1.84%	0.5059	2.07%	0.6832	8.23%	0.3885	4.56%	0.7855	4.08%	0.4733	4.36%
Baseline + CH	0.8031	0.92%	0.5003	1.51%	0.6318	3.09%	0.3692	2.63%	0.7629	1.82%	0.4395	0.98%
Baseline + SP	0.8065	1.26%	0.5017	1.65%	0.6491	4.82%	0.3572	1.43%	0.7776	3.29%	0.4547	2.50%
Complete - FC	0.8112	0.38%	**0.5064**	0.27%	**0.6877**	0.41%	**0.3899**	1.03%	**0.7942**	1.36%	**0.4751**	0.49%
					ISS							
	Pre	Δ	Suc	Δ	Pre	Δ	Suc	Δ	Pre	Δ	Suc	Δ
Baseline + SA	0.8313	5.46%	0.5427	2.94%	0.6227	2.56%	0.3316	2.81%	0.7907	3.76%	0.4881	4.63%
Baseline + CA	**0.8419**	6.52%	0.5462	3.29%	0.6249	2.78%	0.3435	4.00%	0.7921	3.90%	0.4899	4.81%
Baseline + CH	0.8266	4.99%	0.5385	2.52%	0.6241	2.70%	0.3319	2.84%	0.7857	3.26%	0.4870	4.52%
Baseline + SP	0.8331	5.64%	0.5422	2.89%	0.6217	2.46%	0.3321	2.86%	0.7897	3.66%	0.4908	4.90%
Complete - FC	0.8412	0.95%	**0.5487**	0.37%	**0.6301**	0.54%	**0.3489**	1.09%	**0.7933**	0.33%	**0.4911**	0.84%

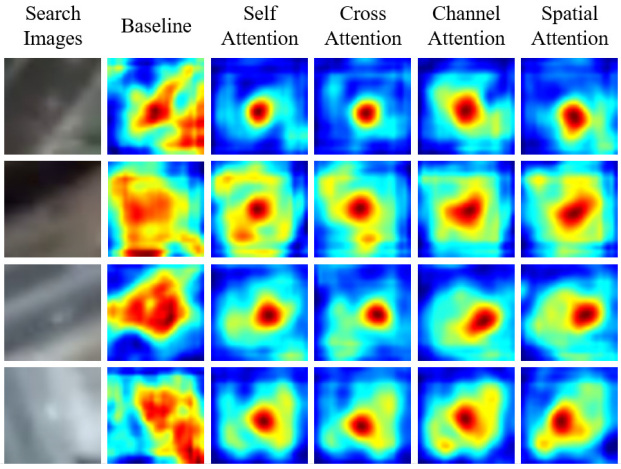


Fig. 13. Visualization of the effectiveness of our attention modules. The 1st column: search images, the 2nd column is the intermediate feature map of the baseline model, and the 3rd–6th columns are the intermediate map of the four attention modules, respectively. The location of the feature map used for visualization is the input of the RPN.

F. Ablation Experiments

In this section, we will validate the key components of the proposed tracker to see how they contribute to performance improvement. The results of ablation experiments are summarized in Table IV, and the visualization results of feature layers are shown in Fig. 13.

Specifically, in Table IV, the ablation experiments on JSANet are divided into five parts or to say five lines by the impact of self-attention (Baseline + SA), cross-attention (Baseline + CA), channel attention (Baseline + CH), spatial attention (Baseline + SP), and feature combination (Complete -FC), and each part consists of three subparts of overall (All), SI, and BCs. The Pre and Suc in the table represent precision and success, respectively. Δ represents the percentage difference, and there are a series of colored percentages below Δ. The red percentages represent performance improvement after the module was added to the baseline model SiamRPN++, while the blue ones represent the performance decline of our proposed JSANet after the feature fusion module was removed.

1) Visualization: Fig. 13 shows the heat map comparison of before and after embedding our modules into the baseline model. It can be seen that before applying our attention modules, the feature maps extracted from backbone may fail to focus on the foreground objects when facing weak feature targets or environmental interference (e.g., Line 2). In addition, even if the target position can be locked, the thermal value of the background area around the target may fail to converge, affecting the size regression of the boundary box (e.g., Line 1). In contrast, our model not only accurately matches the target position at the feature peak points but also effectively suppresses the surrounding background information to avoid abnormal deformation of boundary frames. This visualization well demonstrates the effectiveness of the proposed attention modules.

2) Self-Attention: The design idea of the self-attention approach makes the model become an "enhanced version" of SiamRPN++. The two branches interact with themselves, improving the ability of feature screening and focusing from the channel and spatial views. As shown in the first line of the Table IV, the performance of the algorithm has been relatively obviously improved. This is exactly the benefit from the fine utilization of extracted features through the self-attention mechanism.

In addition to comparison with the baseline model, comparison with classical self-attention models represented by Nonlocal and Triplet will be more convincing. In Table V, the experimental results of three self-attention models and baseline model SiamRPN++ are collected intensively. As can be seen,

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TABLE V
PERFORMANCE COMPARISON AMONG SA, TRIPLET, AND NONLOCAL

	All		SI		BC	
Method	Pre	Suc	Pre	Suc	Pre	Suc
Jilin-1						
Baseline + SA	**0.8075**	**0.5053**	**0.6656**	**0.3759**	0.7701	**0.4659**
Triplet	0.7976	0.5026	0.6388	0.3691	0.7558	0.4615
Non-local	0.8008	0.4978	0.6421	0.3663	**0.7702**	0.4586
Baseline	0.7939	0.4852	0.6009	0.3429	0.7447	0.4297
ISS						
Baseline + SA	**0.8313**	**0.5427**	0.6227	**0.3316**	**0.7907**	0.4881
Triplet	0.8253	0.5394	**0.6277**	0.3298	0.7886	**0.4885**
Non-local	0.8192	0.5339	0.6105	0.3254	0.7842	0.4869
Baseline	0.7767	0.5133	0.5971	0.3035	0.7531	0.4418

TABLE VI
INFERENCE SPEED FOR 16 TRACKERS

Method	FPS	Method	FPS	Method	FPS
		JSANet	33.10		
Triplet	32.17	TrDiMP	16.87	KYS	15.17
Non-local	38.58	ECO	32.73	PrDiMP	15.67
SiamRPN++	54.39	KCF	**95.61**	SiamBAN	48.41
TrTrOnline	21.34	SuperDiMP	14.55	C-COT	26.61
RT MDNet	20.47	SiamCAR	18.09	SiamDW	68.27

the performance of the self-attention model is significantly improved after embedding the self-attention model into the baseline model, and the performance of self-attention proposed by us is not weaker than that of the classical model, proving its effectiveness.

3) Cross-Attention: The idea of cross-attention is similar to SiamRPN++ using SiamRPN in multiple layers. Both of them hope to increase the opportunity of feature interaction between two branches as much as possible, through cascade fusion of multilevel interaction results improving the discrimination ability of the model. Merely cross-attention stands from the perspective of features but baseline from the perspective of regression. For weak feature remote sensing targets, interactive learning between the two branches, except focusing on the key areas, also enriches its feature with the guidance information of the other side. As shown in the second line of the Table IV, the algorithm achieves significant performance improvement, which benefits from the good coordination between cross-attention and baseline models.

4) Channel and Spatial Attention: The single attention mechanism lacks procedures for mutual supervision and joint optimization between the channel and spatial. For pure channel attention, especially facing weak feature targets, the lack of spatial supervision may lead to improper screening and the following process of the effect model. Conservative screening is difficult to improve performance, and excessive screening may lead to feature loss causing the opposite effect. For pure spatial attention, the noise features contained in the target are likely to mislead the definition of key areas, further misleading bounding box regression. The results in the third and fourth lines of the Table IV support the above views. The performance of pure channel and spatial attention as a whole is lower than self- and cross-attention and shows instability.

5) Feature Combination: The design idea of the feature combination strategy is to introduce adaptive parameters to balance the weight of each attention further improving the fusion effect. For JSANet discarding feature combination, the output of each module is directly added. From the experimental results shown in the last line of Table IV, the result of direct addition is not ideal, and even backward instead of forward (compared with an overall precision of cross-attention). In contrast, comparative results prove that the integration strategy could achieve a stable performance improvement, which proves that it is necessary to design and adopt a reasonably designed fusion strategy when pursuing the ultimate performance of the model.

G. Inference Speed Evaluation

To test the reasoning speed of each algorithm, we randomly selected five times from a total of 522 targets in ten videos, and 30 targets were selected each time to form a mini dataset (the targets would be put back into the random pool after being selected for statistics). Each algorithm then tests these datasets and calculates the FPS, which is averaged to get the reasoning speed of these algorithms. The GPU device used in our test is NVIDIA Tesla V100, and the test results are shown in Table VI.

After analysis, it is found that the efficiency of an algorithm is judged not only by the number of its parameters but also by its "operation logic." For example, KYS runs low because it contains many postprocessing strategies. Similarly, some algorithms also have some built-in guarantee strategies, such as expanding the search range when the target confidence is lower than the set threshold, which will also lead to an "implicit relationship" between the tracking efficiency of the algorithm and the difficulty of the target.

V. CONCLUSION

In this article, a JSANet is proposed for vehicle object tracking in satellite videos. We introduce a Siamese attention mechanism consisting of both self-attention and cross-attention modules, which can be further divided into channel and spatial submodules. Through joint optimization of each module, implementing information screening, and key areas' focusing, the new Siamese attention can strongly enhance the ability of the model to identify the targets, and at the same time improve the robustness against SI and BCs. In addition, a self-adaptive feature combination strategy is designed to further increase the tracking accuracy. Extensive experiments are conducted on the Jilin-1 and ISS satellite videos, and our

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JSANet achieved the state-of-the-art results compared with 15 excellent trackers.

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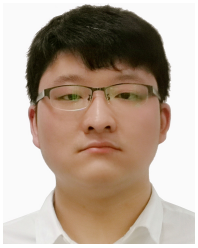
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